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Mark schemes

Q1.

(a) Peg is smooth

B1: Correct assumption

B1

1

(b) String is light

B1: First correct assumption

B1

String is inextensible or inelastic

B1: Second correct assumption

B1

Tension is the same throughout the string

Note: Ignore any additional assumptions.

2

(c) $11g - T = 11a$

M1: Equation of motion for A, containing, $11g$ or 107.8 and $11a$.

M1

A1: Correct equation

A1

$T - 9g = 9a$

M1: Equation of motion for B containing, $9g$ or 88.2 and $9a$.

M1

A1: Correct equation

A1

$2g = 20a$

AG

$a = 0.98$ m s^{-2}

A1: Correct acceleration from correct orking.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give

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their final answer as 0.98. If final answer is – 0.98 don't award final A1 mark.

Special Case:

Whole String Method $2g = 20a$ and

$a = 2g/20 = 0.98$ OE M1A1A1

Use of $g = 9.81$ gives 0.981. If this is the first time award M1A1M1A1A0, but don't penalise again on the same script.

5

(d) (i) $v = 0 + 0.98 \times 0.5 = 0.49$ m s^{-1}

M1: Use of constant acceleration equation to find v with $u = 0$, $a = 0.98$ and $t = 0.5$.

M1

A1: Correct v

A1

2

(ii) $s = 0 + \frac{1}{2} \times 0.98 \times 0.5^2 = 0.1225$ m

M1: Finding distance travelled by each particle with $u = 0$, $a = 0.98$ and $t = 0.5$.

M1

A1: Correct distance. Accept 0.122 or 0.123

A1

OR

$0.49^2 = 0^2 + 2 \times 0.98s$

M1: Finding distance travelled by each particle with $u = 0$, $a = 0.98$ and their v

(M1)

$$s = \frac{0.49^2}{2 \times 0.98} = 0.1225$$

A1: Correct distance. Accept 0.122 or 0.123

(A1)

$d = 2 \times 0.1225$

M1: Doubling distance or use of $d/2$ in their original equation.

M1

= 0.245 m

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A1: Correct final distance. Allow 0.244 or 0.246.
(Use of 0.5 × 0.49 = 0.245 scores zero
unless justified)

A1

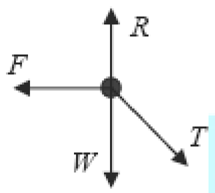
4

If candidates calculate the distance first
award marks as above (see (d) (i)) or:
M1: Use of constant acceleration equation
to find v with u = 0, a = 0.98 and s = 0.1225.
A1: Correct v
Note: If parts (i) and (ii) are not
separated or clearly labelled still award
marks for both parts if justified.

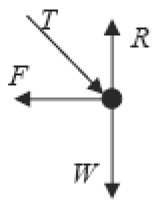
[14]

Q2.

(a)



OR



B1: Diagram with four forces showing arrow heads and labelled.
Allow mg or 8g.
Allow T or 40 or other reasonable notation.
Allow μR.
Direction of friction must be to the left.
Any components must be shown in a different style.

B1

1

(b) 8g + 40 sin 30° (= R)

M1: Expression for normal reaction, with
mg or 8g and 40 sin 30° or 40 cos 30°.
Allow incorrect signs.

M1

A1: Correct expression with correct signs.

A1

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(R =) 98.4 N **AG**

A1: Correct value from correct working.
Use of g = 9.81 gives 98.5 N. Do not enalise if you have already done so arlier in the scripts. Otherwise penalise by 1 mark.

A1

3

(c) $F = 40 \cos 30° = 34.6$ N

M1: Use of 40 cos 30° or 40 sin 30°.
Award M0 if any extra terms

M1

A1: Correct value for friction. Don't need to see F.

A1

2

(d) $40 \cos 30° \leq \mu \times 98.4$

M1: Use or $F \leq \mu R$ (or $F = \mu R$). Must use R = 98.4 and a positive value for F.

M1

A1F: Correct inequality or equation
Allow use of $F = \mu R$ throughout.

A1F

$$\mu \geq \frac{40\cos 30°}{98.4}$$

$$\mu \geq 0.352$$

A1F: Correct minimum value. For follow through must use R = 98.4 and their value for F from part (c). For example use of sin 30° in part (c) gives 0.203.

A1F

3

[9]

Q3.

(a) $13^2 = 0^2 + 2 \times 1.3s$

M1: Use of a constant acceleration equation to find distance.

M1

A1: Correct equation

A1

$$s = \frac{13^2}{2.6} = 65 \text{ m}$$

A1: Correct distance

A1

3

(b) (i) $3900 - 800 - P = 2000 \times 1.3$

M1: Four term equation of motion for car and trailer.

M1

A1: Correct equation

A1

$P = 3900 - 800 - 2600 = 500$ N

A1: Correct value for P

A1

3

(ii) $T - 500 = 600 \times 1.3$

M1: Three term equation of motion for trailer.

M1

A1: Correct equation

A1F

$T = 500 + 780 = 1280$ N

A1: Correct tension

A1F

3

[9]

Q4.

(a) $R = 14 \times 9.8 = (137.2)$

M1: Finding the normal reaction. Accept 14g.

M1

$F = 0.25 \times 137.2$ **OR** $F = 0.25 \times 14 \times 9.8$

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M1: Use of $F = \mu R$

M1

$F = 34.3$ N

A1: Correct friction
Use of $g = 9.81$ gives
$R = 137.3$ and $F = 34.3$ so in this case do not penalise use of $g = 9.81$.

A1

3

(b) $6g - T = 6a$

M1: Equation of motion for the particle, containing T, $6g$ or 58.8 and $6a$.
A1: Correct equation with correct signs.

M1 A1

$T - 34.3 = 14a$

M1: Equation of motion for the block, containing T, 34.3 or their F and $14a$.
A1: Correct equation with correct signs.

M1 A1

$6g - 34.3 = 20a$

$a = \dfrac{6g - 34.3}{20} = 1.225 \text{ ms}^{-2}$ AG

A1: Correct acceleration from correct working.
If – 1.225 is obtained from consistent working award 4 marks and if changed to +1.225 with an explanation, award full marks.
Special Case:
Whole string method
$6g - 34.3 = 20a$
$a = 1.225$ OE
award M1A1A1
Use of $g = 9.81$ gives
$a = 1.228$ penalise use of
$g = 9.81$ by deducting 1 mark, but don't penalise again on the same script.

(c) $T - 34.3 = 14 \times 1.225$

M1: Use of either of candidates equations of motion to find tension, with $a = \pm 1.225$ and their F (Method 1).

M1

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$T = 17.15 + 34.3 = 51.5$ N

A1: Correct tension
Accept 51.45 or 51.4. Don't penalise use of g = 9.81 if already done in part (b).

A1

OR

$6g - T = 6 \times 1.225$

(M1)

$T = 6 \times 9.8 - 6 \times 1.225 = 51.5$

(A1)

2

(d) $v^2 = 0^2 + 2 \times 1.225 \times 0.8$

M1: Use of constant acceleration equation to find speed with u = 0.
A1: Correct equation

M1A1

$v = \sqrt{1.96} = 1.4 \text{ ms}^{-1}$

A1: Correct speed AWRT 1.4

A1

OR

$0.8 = \frac{1}{2} \times 1.225t^2$

In method 2, no marks awarded for just finding t.

$t = (1.1428)$

(M1)

$v = 1.225 \times 1.1428$

(A1)

$= 1.40$

(A1)

3

(e) $v^2 = 1.4^2 + 2 \times 9.8 \times 0.5$

M1: Use of constant acceleration equation

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to find speed with u = 1.4 or their answer to part (d), a = ±9.8 and s = 0.5.
A1F: Correct equation.
Follow through their answer to part (d).

M1A1F

$v = 3.43\ \text{m s}^{-1}$

A1F: Correct speed.
Don't penalise use of g = 9.81 if already done earlier in question.
In method 2, no marks awarded for just finding t.

A1F

OR

$0.5 = 1.4t + 4.9t^2$

$t = 0.2071$

$v = 1.4 + 9.8 \times 0.2071$

(M1)(A1F)

= 3.43 m s^{-1}

(A1F)

3

[16]

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Q5.

(a) (i) 20 × 9.8 = R + 60 sin 30°

M1: Equation or expression for normal reaction with mg or $20g$ or 196 and 60 sin 30° or 60 cos 30°.
A1: Correct equation or expression with correct signs.

M1A1

(R =) 20 × 9.8 – 60sin30° = 166 N AG

A1: Correct value from correct working.
Must be positive.
Don't penalise use of g = 9.81 if already done earlier on script. Should still get 166, but from 166.2.

A1

3

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(ii) $166\mu = 60 \cos 30°$

M1: Use of $F = \mu R$, with R = 166 or 166.2. Do not allow inequalities here.
M1: Resolving horizontally with cos30° or sin30° oe
A1: Correct equation
Examples:
$166\mu = 60$ M1M0A0
$166\mu = -60 \cos 30°$ M1M1A0

M1M1A1

$$\mu = \frac{60\cos 30°}{166}$$
$= 0.313$

A1: Correct coefficient of friction.

A1

4

(b) $20 \times 0.8 = T \cos 30° - 0.313(20 \times 9.8 - T \sin 30°)$

B1: $20g - T \sin 30°$ oe seen.
M1: Three term equation of motion, where normal reaction is dependent on T.
A1F: Correct equation

B1M1A1F

$$T = \frac{20 \times 0.8 + 0.313 \times 20 \times 9.8}{\cos 30° + 0.313 \sin 30°} = 75.6\text{ N}$$

dM1: Solving for T including factorisation.
A1F: Correct tension.
AWRT 75.6
Follow through incorrect values of μ from part (a).
Don't penalise use of g = 9.81 if already done earlier on script. Should get 75.7.
Allow 75.8 if intermediate values rounded.

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5

[12]

Q6.

(a) $R = 15 \times 9.8 = 147$ N
$F = 0.2 \times 147 = 29.4$ N

Finding normal reaction.

M1

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Use of $F = \mu R$

M1

Correct friction force.

A1

3

(b) $5g - T = 5a$

Three term equation of motion for 5 kg particle

M1

Correct equation

A1

$T - 29.4 = 15a$ **AG**

Three term equation of motion for 15 kg block

M1

$49 - 29.4 = 20a$

Correct equation

A1

$a = 0.98$ m s^{-2}

Correct final answer from correct working.

A1

5

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(c) $1.4^2 = 0^2 + 2 \times 0.98s$

$s = \frac{1.4^2}{1.96} = 1\text{m}$

Use of a constant acceleration equation with $u = 0$

M1

Correct equation

A1

Correct distance

A1

3

[11]

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Q7.

(a)

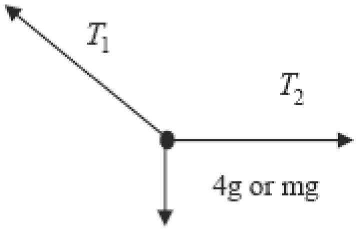

Diagram with three forces, labels and arrow heads. Different variables must be used for each tension

B1

1

(b) T_1 sin 30° = 4 × 9.8

Two term equation from resolving vertically. Must see a sin or cos term for M1

M1

$T_1 = \frac{4 \times 9.8}{\sin 30°} = 78.4$ N **AG**

Correct equation

A1

Correct tension from correct working.

A1

3

(c) T_2 = 78.4 cos 30° = 67.9 N

Two term equation from resolving horizontally.

M1

Correct tension.

A1

2

[6]

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Q8.

(a) (i) 0.2a = –0.2 × 9.8 sin 20°

AG

a = –9.8 sin 20° = – 3.35 m s^{-2}

Two term equation of motion with weight resolved

M1

Correct equation

A1

Correct acceleration from correct working

A1

SC No negative sign but otherwise correct award M1A1A0
Allow $a = -g \sin 20°$

3

(ii) $0 = 4^2 + 2 \times (-3.35)s$

Use of constant acceleration equation with $v = 0$ and $u = 4$

M1

Correct equation

A1

$s = \frac{16}{6.7} = 2.39\text{ m}$

Correct distance

A1

3

(iii) The puck slides back down the slope as the puck is at rest and the resultant force is now acting down the slope / no friction / smooth slope.

Slides back down

B1

Acceptable explanation

E1

2

(b) (i) $R = 0.2 \times 9.8 \cos 20°$ **AG**

Finding normal reaction by resolving. Must see a trig term.

M1

$F = 0.5 \times 0.2 \times 9.8 \cos 20°$

Use of $F = \mu R$

M1

= 0.921 N

Correct friction from correct working.

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A1

3

(ii) $0.2a = -0.921 - 0.2 \times 9.8 \sin 20°$

Three term equation of motion with the weight resolved

M1

Correct equation

A1

$a = -7.96$ m s^{-2}

Correct acceleration (with or without the minus sign, applied to both A1 marks)

A1

3

(iii) The puck stays at rest because the friction has a maximum of 0.921 and the component of the weight down the slope is less (0.670)

Stays at rest

B1

Acceptable explanation

dE1

2

[16]

Q9.

(a) $F = 0.4 \times 1000 \times 9.8$

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Use of $F = \mu R$

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M1

= 3920 **AG**

Correct friction from correct working.
Allow $F = 0.4 \times 9800$
Allow verification

A1

2

(b) $P - 3920 = 5000 \times 0.8$

Three term equation of motion including an explicit 0.8

M1

Correct equation

A1

P = 7920 N **AG**

Correct force from correct working.
Allow P = 5000 × 0.8 + 3920

A1

3

(c) T – 3920 = 1000 × 0.8

Three term equation of motion

M1

Correct equation

A1

T = 4720 N

Correct tension

A1

or
7920 – T = 4000 × 0.8
T = 4720 N

3

(d) Friction is reduced because the normal reaction is reduced.

Friction reduced

B1

Acceptable explanation

E1

2

[10]

Q10.

(a) 2 × 9.8 – T = 2 × 0.9

Three term equation of motion for B

M1

Correct equation

A1

T = 19.6 – 1.8 = 17.8 N

Correct tension

A1

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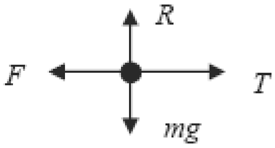
3

(b)



Correct force diagram with labels and arrows

B1

1

(c) $R = 3 \times 9.8 = 29.4$ N

Calculation of normal reaction force.

B1

1

(d) $17.8 - F = 3 \times 0.9$

Three term equation of motion for A.

M1

$F = 17.8 - 2.7 = 15.1$ N

Correct friction

A1

2

(e) $15.1 = 29.4\mu$

Use of $F = \mu R$

M1

$\mu = \frac{15.1}{29.4} = 0.514$

Correct value for μ

A1

2

[9]

Q11.

(a) $\mathbf{F} = 5\mathbf{j} + 8\mathbf{i} - 7\mathbf{j} = 8\mathbf{i} - 2\mathbf{j}$

Adding the two forces. For incorrect answers, evidence of adding must be seen

M1

Correct resultant

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A1

2

(b) $F = \sqrt{8^2 + 2^2} = \sqrt{68} = 8.25$ N

Finding magnitude (must see addition and not subtraction)

M1

Correct magnitude

Accept $2\sqrt{17}, \sqrt{68}$ *or AWRT 8.25 (eg 8.246)*

A1F

2

(c)

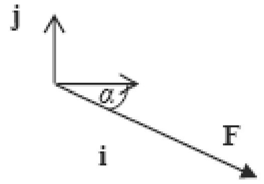

Diagram with force in the correct quadrant and with correct direction shown by an arrow.

B1

$\tan \alpha = \frac{2}{8}$

Using trig to find angle: if tan, 8 in denominator; if sin or cos, 8.25 or their answer to part (b) in denominator

M1

$\alpha = 14.0°$

Correct angle
Accept 14.1 or 14 or AWRT 14.0 (eg 14.04)
M1 and A1 not dependent on B1

A1

3

[7]

Q12.

(a) (i) $T = 6 \times 9.8 = 58.8$ N

Use of tension being equal to the weight
Accept 6g

B1

1

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(ii) $58.8 = T + 4 \times 9.8$

Three term equation for equilibrium containing 58.8, T and 4×9.8 or equivalent terms.
For M1, 58.8 can be replaced by candidates answer to part (a)(i) provided it is not zero.

M1

Correct equation

A1

$T = 58.8 - 39.2$
$= 19.6$ N

Correct tension
Accept $2g$

A1

3

(b) $6g - T = 6a$

Three term equation of motion for 6 kg particle containing 58.8 or $6g$, T and $6a$.

M1

Correct equation

A1

$T - 4g = 4a$

Three term equation of motion for 4 kg particle containing 39.2 or $4g$, T and $4a$.

M1

Correct equation

A1

$2g = 10a$
$a = 1.96$ m s^{-2}

Correct acceleration
Candidates who work consistently to obtain $a = -1.96$ gain full marks.

A1

5

Special Case for whole system
$6g - 4g = 10a$

Difference in weights equal to $10a$

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(M1)

A1: Correct equation

(A1)

a = 1.96

A1: Correct acceleration

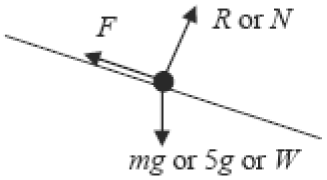
(A1)

(3)

[9]

Q13.

(a)



Correct force diagram with labels and arrows
Accept components of the weight if shown in a different notation with the weight also shown.
B0 if components are shown instead of the weight.

B1

1

(b) (R =) 5 × 9.8cos40° = 37.5 N **AG**

Attempt at resolving perpendicular to the slope (eg 49sin40°)

M1

Correct value from correct working

A1

2

(c) 5 × 0.8 = 5 × 9.8sin40° – μ × 5 × 9.8cos40°

Use of $F = \mu R$ at any stage and with any F but with R = 37.5 OE

B1

Three term equation of motion from resolving parallel to the slope with weight component, friction and ma term.

M1

Correct terms seen (may be as 31.5, 37.5μ (or F) and 4)

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A1

Correct signs

A1

$$\mu = \frac{5 \times 9.8 \sin 40° - 5 \times 0.8}{5 \times 9.8 \cos 40°} = 0.733$$

Solving for μ

m1

A1: Correct value for μ

Allow 0.732 but not $\frac{11}{15}$ unless converted to a decimal

A1

6

(d) There is less friction so the coefficient of friction must be less.

Less friction

B1

Smaller coefficient of friction
If the answer and explanation contradict each other, award no marks

B1

2

[11]

Q14.

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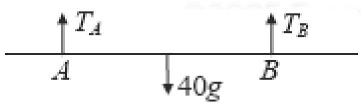
(a)



B1

1

(b) Taking moments about A
$2.1 \times 40g = T_B \times 4$

B1 for 2.1

M1 B1

$T_B = 21g$

A1

3

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(c) Resolve vertically $T_A + T_B = 40g$

M1

$T_A = 19g$ or 186 N

A1

2

(d) Gravitational force acts through mid point of the rod

E1

1

[7]

Q15.

(a) 900 × 0.8 = T – 800

Three term equation of motion for the caravan

M1

Correct equation

A1

T = 720 + 800 = 1520 N

Correct result from correct working; AG

A1

3

(b) 2400 × 0.8 = F − 400 − 800

Four term equation of motion for the combined body or car

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M1

Correct equation

A1

F = 1920 + 1200 = 3120 N

Correct force

A1

3

[6]

Q16.

(a) $F = \sqrt{6^2 + 5^2}$

Obtaining an equation for F with square

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or root. Correct equation

M1A1

$= \sqrt{61} = 7.81$

Correct force

A1

Alt

$\alpha = \tan^{-1}\left(\frac{5}{6}\right) = 39.8°$

$F = \frac{6}{\cos 39.8} = 7.81$ or

Equation for F with a value for α. Correct equation

(M1A1)

$F = \frac{5}{\sin 39.8} = 7.81$

Correct force

(A1)

3

(b) $\alpha = \tan^{-1}\left(\frac{5}{6}\right)$ or $\cos^{-1}\left(\frac{6}{7.81}\right)$ or $\sin^{-1}\left(\frac{6}{7.81}\right)$

Obtaining an equation for α using trigonometry. Correct equation (using their F)

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M1A1

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= 39.8°

Correct angle

A1

Accept values between 39.7 and 39.9

Alt

$\frac{\sin\alpha}{5} = \frac{\sin 90°}{\sqrt{61}}$

$\alpha = 39.8°$

3

[6]

Q17.

(a) The string is light and inextensible or inelastic or taut

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First assumption

B1

Second assumption

B1

2

(b) 6 = 0 + 4a

Finding a using a CA equation

M1

$a = \frac{6}{4} = 1.5$

Correct a from correct working

A1

2

(c) 7 × 9.8 – T = 7 × 1.5

Three term equation of motion with F for the 7 kg particle. Correct equation

M1A1

T = 68.6 – 10.5 = 58.1

Correct tension

A1

3

(d) 58.1 – F = 13 × 1.5

Three term equation of motion with F for the 13 kg particle. Correct equation

M1A1

F = 58.1 – 19.5 = 38.6

Correct F

A1

R = 13.98 = 127.4

Correct R

B1

38.6 = μ × 127.4

Use of $F = \mu R$

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dM1

$\mu = \dfrac{38.6}{127.4} = 0.303$

Correct coefficient of friction

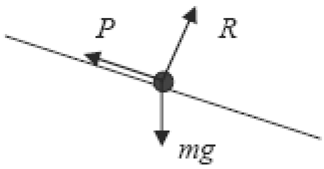
A1

6

[13]

Q18.

(a)



Correct diagram with arrows and labels
Must not use F instead of P
Condone resistance instead of P

B1

1

(b) P = 100 × 9.8 sin 4°

Resolving weight (must see 100)

M1

Using sin 4° or cos 86°

M1

= 68.4

AG *Correct P from correct working*

A1

3

(c) 100a = 100 × 9.8 sin 5° − 100 × 9.8 sin 4°

Three term equation of motion

M1

Weight resolved correctly

A1

Correct equation

A1

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$$a = \frac{100 \times 9.8 \sin 5° - 100 \times 9.8 \sin 4°}{100}$$

= 0.171

Correct a. (Accept 0.170 or 0.17)

A1

4

(d) You would expect P to vary with the speed of the car.

Correct explanation

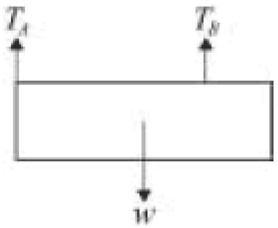
B1

1

[9]

Q19.

(a)



Arrows + labels, w in centre

B1

1

(b) M(A) $0.4W = 0.6T_B$

Moments equation

M1

$$T_B = \frac{2W}{3}$$

Accept 2 dp for each A1

A1

Res ↑ or M(B) $T_A = \frac{W}{3}$

M1A1

4

(c) Lamina is uniform $\Rightarrow$ weight acts at centre

B1

1

[6]

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Q20.

(a) $320 = \frac{1}{2} \times a \times 80^2$

Use of constant acceleration equation with $u = 0$

M1

Correct equation

A1

$a = \frac{2 \times 320}{80^2} = 0.1\,\text{m s}^{-2}$

AG Correct acceleration from correct working

A1

3

(b) $v = 0 + 0.1 \times 80$

Use of constant acceleration equation with $u = 0$

M1

$= 8\ \text{m s}^{-1}$

Correct velocity

A1

2

(c) $L - 450 \times 9.8 = 450 \times 0.1$

Three term equation of motion

M1

Correct equation

A1

$L = 45 + 4410$
$= 4455$
$= 4500$ N (to 3sf)

AG Correct force

A1

3

(d) 4410 N

Correct force

B1

1

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[9]

Q21.

(a) $3.45g - T = 3.45a$

Three term equation of motion for one particle

M1

Correct equation

A1

$T - 1.45g = 1.45a$

Three term equation of motion for other particle

M1

Correct equation

A1

$2g = 4.9a$

$a = \frac{2 \times 9.8}{4.9} = 4\ \text{m s}^{-2}$

AG Correct acceleration from correct working

A1

5

(b) $T = 1.45 \times 4 + 1.45 \times 9.8$

Use of one equation from (a) to find T

M1

= 20.01
= 20.0 N (to 3 sf)

Correct T

A1

2

(c) $s = \frac{1}{2} = 0.5\ \text{m}$

Use of $s = \frac{1}{2}$

M1

$v^2 = 2 \times 4 \times 0.5$

Use of constant acceleration equation with

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$u = 0$

M1

$v = 2\ \text{m s}^{-1}$

Correct speed
(no negative sign)

A1

3

[10]

Q22.

(a) $T_1 \sin 35° = T_2 \sin 35°$

Resolving two forces and forming an equation, with different tensions for each string

M1

$T_1 = T_2$

Correct result from correct working

A1

OR

$T_1 \cos 55° = T_2 \cos 55°$

$T_1 = T_2$

2

(b) $T_1 \cos 35° + T_2 \cos 35° = 2 \times 9.8$

$T_1 \cos 35° + T_1 \cos 35° = 2 \times 9.8$

Resolving forces to form a three term vertical equation

M1

Correct equation

A1

T_1 or T_2 eliminated correctly

A1

$T_1 = \dfrac{2 \times 9.8}{2\cos 35°} = 12.0$ N (to 3sf)

Solving for T_1 or T_2

dM1

Correct tension
Accept 12 N or 11.9 N

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A1

5

(c) $2 \times 40 \cos 35° = 9.8m$

Forming an equation with two tensions to find m

M1

Correct equation

A1

$$m = \frac{80\cos 35°}{9.8} = 6.69\,\text{kg}$$

Correct mass
Acept 6.68

A1

OR

$$m = \frac{40}{11.96} \times 2$$

(M1)(A1)

= 6.69 kg

(A1)

3

[10]

Q23.

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(a) $T - 800 = 1200 \times 0.4$

Three term equation of motion for the car

M1

Correct equation

A1

$T = 800 + 480$
$= 1280$ N

Correct tension

A1

Treat calculation of two tensions as two methods unless one selected
Treat sum or difference of two tensions as an incorrect method

3

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(b) $3000 - 800 - F = 4000 \times 0.4$

Four term equation of motion (truck or both)

M1

Correct terms

A1

Correct signs

A1

$F = 3000 - 800 - 1600$
$F = 600$ N

AG Correct resistance force from correct working

A1

OR
$3000 - 1280 - F = 2800 \times 0.4$
$F = 3000 - 1280 - 1120$
$F = 600$ N

4

(c) Increase, because a greater tension would be needed so that the horizontal component would be the same as the tension above.

Greater

B1

Reason
Second B1 dependent on the first B1 mark

B1

2

[9]

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Q24.

Moments about A:

OE

$6S = 4 \times 30g$

M1A1

$S = 20g$ N or 196 N

A1

Resolving vertically: $R + S = 30g$

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M1

R = 10g N or 98 N

SC3 20 N; 10 N

A1

5

[5]

Q25.

(a) Centre of mass of rod is 3 m from river bank

Use of centre of mass is centre of rod

B1

Taking moments about A, edge of bank:

3 × 15 = 50x

M1

x = 0.9

A1

Or resolve R = 65g B1

Moments about any point (correct) M1

0.9 A1

3

(b) Taking moments about A:

50 × 2 = 15 × 3 + m × 8

M1 3 terms, 2 correct

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M1A1

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55 = 8m

A1

$m = 6\frac{7}{8}$

Mass is $6\frac{7}{8}$ kg

Accept 6.88 and 6.87

A1

4

(c) Centre of mass of rod is 3 m from river bank

Centre of mass is at centre of rod

E1

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1

(d) eg Woman is a particle
The mass is a particle
The plank is a rigid rod

E1
1

[9]

Q26.

(a) (i) T = 0.6 × 9.8 = 5.88N or 0.6g

B1
1

(ii) Force = 2T = ↓ 11.76N or 11.8N or 1.2g

B1
2

Magnitude

B1

direction

B1

(b) (i) Q:$0.8g$ – T = $0.8a$

M1

T – $0.6g = 0.6a$
either equation

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A1

$0.2g = 1.4a$

A1

alt *M1A1 if solving for T, full method for solving,*
A1 accurate attempt

m1

$a = 1.4a$

A1

T = 6.72N
CAO
S.C *whole string.*
To find a: 0.2g = 1.4a (M1)

a = 1.4 A1
To find T: *(M1A1)*
CAO

A1 **6**

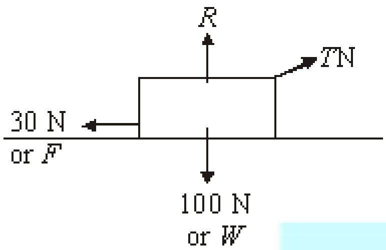
(ii) Force = 2T = 13.44N

B1 **1**

[10]

Q27.

(a)



B1 **1**

(b) F = μR, 30 = 0.5 R

M1

R = 60N

A1 **2**

(c) Constant speed $\Rightarrow$ no acceleration / forces

M1

balance $\therefore$ 30N = T_x

A1 **2**

(d) T_y + R = 100

M1

T_y = 40

A1

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$T = \sqrt{(30^2 + 40^2)}$

Use of a vertical component of T

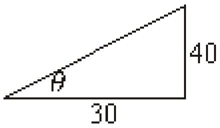
M1

$T = 50$

A1

4

(e)



$\tan\theta = \frac{40}{30}$

M1

A1ft

$\theta = 53.1°$ Accept 53°

CAO

A1

3

[12]

Q28.

(a) (i) $R = 80\cos 25°$

component attempted

M1

correct component

A1

$R = 72.5\text{N}$

CAO

A1

3

(ii) $F = 0.32 \times 72.5$

condone inequality

M1

$F = 23.2\text{N}$

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CAO

A1
2

(iii) $T + F = 80\cos 65°$
3 forces direction correct, component

M2

attempted

A1

$T = 10.6$N
ft friction

A1ft
4

(iv) $T = F + 80\cos 65°$
3 forces, direction correct, component

M1

attempted

A1

$T = 57.0$N (57N)
ft friction

A1ft
3

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(iv) Mass = $\frac{80}{g}$ = (8.16kg)

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B1
1

(b) $80 \cos 65° - F =$ mass $\times$ acceleration
3 terms, component attempted

M1

$10.6 = \frac{80}{g} \times$ acc
all correct

A1

acc = 1.30 ms^{-2} (1.3 ms^{-2})
CAO

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A1
3
[16]

Q29.

$5T_A = 20 \times 9.8 \times 1.5$

Moment equation.

M1

Correct equation

A1

$$T_A = \frac{20 \times 9.8 \times 1.5}{5} = 58.8 \text{ N}$$

Correct tension

A1

Vertical equation with T or moments equation.

M1

$T + 58.8 = 20 \times 9.8$

Correct equation

A1

$T = 137.2\text{N}$

Correct tension

A1
[6]

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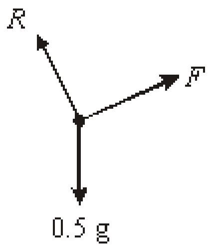
Q30.

(a)



Accept W or mg (or 4.9) for weight
Arrows and labels needed
Can replace W with 2 correct components

B1
1

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(b) $F = 0.5g\sin 16°$

component of weight attempted

M1

All correct including signs

A1

$F = 1.35\text{N}$

CAO

A1

3

(c) $R = 0.5g\cos 16°$

component of weight attempted

M1

R = 4.71N

CAO

(b) and (c): For resolving horizontal/vertical M1 for each equation, A1 one equation correct

F A1 R A1

A1

2

(d) $F \le \mu R$

$1.35 \le \mu \times 4.71$

Use of candidate's R, accept equals sign

M1

$\mu \ge 0.287$

CAO inequality required

A1

2

[8]

Q31.

(a) $0.3g - T = 0.3a$

Consistent reversal of signs in both equations 4 marks;

M1A1

$T - 0.2g = 0.2a$

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Reversal of signs in one equation, M1 A1 M1 A0

M1A1

$a = 1.96 \text{ ms}^{-2}$

Sign change needs justification
(Whole string:
Equation, 0.3g – 0.2g = 0.5a M1A1,
a = 1.96 A1) max 3/5

A1

5

(b) $0.1g - R = 0.1 \times 1.96$

all terms

M1

all correct

A1

$R = 0.784 \text{ N}$

ft provided R positive

A1F

3

[8]

Q32.

(a) P = 5 + 8cos60°

Both relevant forces, component of 8N attempted

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M1

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All correct

A1

P = 9

CAO

A1

3

(b) Q = 8cos30°

Component of 8N attempted

M1

Q = 6.93 or $4\sqrt{3}$

AWRT 6.93

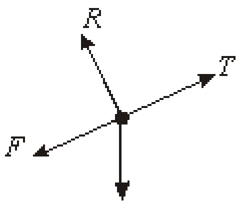
A1

2

[5]

Q33.

(a)



Accept W or mg (or 6.86) for weight
Arrows and labels needed
(can replace W with 2 correct components)

B1

1

(b) $R = 0.7g \cos 22°$

component of weight attempted

M1

$R = 6.36$ N

all correct, including signs

A1

CAO

A1

3

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(c) $F = 0.25 \times 6.36$

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$F = 1.59\,\text{N}$

CAO

M1 A1

2

(d) 5.6 – 0.7g sin 22°

$a = 2.06$ ms^{-2}

4 terms with weight component attempted

M1

A marks -1 each error, accept $\pm 0.7a$

A2

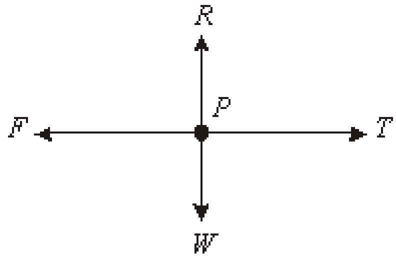
FT one error, accept $\pm$

A1F

[10]

Q34.

(a) (i)



Accept mg, 0.4g or 3.92 for weight
Arrows and labels needed

B1

1

(ii) $F = 0.5 = \times (0.4 \times 9.8)$

Need to see 0.4×9.8 *or 3.92 used*

M1

$F = 1.96$N

A1

2

(b) $T - 1.96 = 1.4a$
$0.3g - T = 0.3a$

M1A1

Consistent reversal of signs in both equations
4 marks; reversal of signs in one equation,
M1 A1 M1 A0

M1A1

$a = 1.4\text{ms}^{-2}$

Sign change needs justification (whole string:
equation, $0.3g - 1.96 = 0.7a$ *M1A1* $a = 1.4$ *A1)*
max 3/5

A1

5

(c) $v = 14 \times 3$
$v = 4.2\text{ms}^{-2}$

Full method

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M1

CAO

A1

2

(d) P : Friction will cause speed to decrease

M1

Accept decelerate or comes to rest

A1

Q : Gravity will cause speed to increase

M1

Accept accelerate

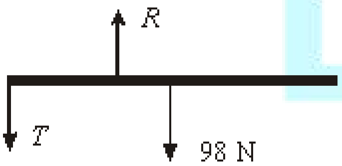
A1

4

[14]

Q35.

(a)



correct force diagram, with labels and arrows.

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B1

1

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(b) $2T = 0.5 \times 98$

moment equation

M1

$T = 24.5$ N

correct equation

A1

AG

correct positive value for the tension from correct working

A1

3

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(c) (i) $2 \times 2 \times 24.5 = 3 \times 9.8 \times m + 0.5 \times 98$

tension doubled

B1

moment equation

M1

$m = \frac{98 - 49}{3 \times 9.8} = \frac{5}{3}$ = 1.67 kg (to 3 sf)

correct equation

A1

correct mass

A1

4

OR

$2 \times 24.5 = 3 \times 9.8m$

for equation

(M1A1)

$m = \frac{49}{29.4} = \frac{5}{3}$ = 1.67 kg

for finding m

(M1A1)

(ii) $R = 24.5 \times 2 + 98 + \frac{5}{3} \times 9.8 = 163\ \text{N}$

considering vertical equilibrium with 3 terms

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M1

correct equation

A1

correct reaction

A1

3

must be consistent with 3(c)(i) if awarding accuracy marks

(d) This allows the centre of mass to be placed at the <u>centre of the rod</u> for the moment calculations.

correct explanation

B1

1

[12]

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